

Quiz: Recursion

Total Score: $/2^2$

Printed Name:

Quiz rules:

1. You MAY use any printed or handwritten notes.
2. You MAY NOT use a computer or any other electronic device.

Problem 1. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     ret = foo(xs[1:])
7     return xs[0] + ret
8 try:
9     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
10 except RuntimeError:
11     print('StackOverflow')
12 EOF
13 $ python3 foo.py
14 $
```

Fraction of LLMs with correct answer: $16 / 16 = 1.00$

Problem 2. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     return foo(xs[1:]) + xs[0]
5 try:
6     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
7 except RuntimeError:
8     print('StackOverflow')
9 EOF
10 $ python3 foo.py
11 $
```

Fraction of LLMs with correct answer: $16 / 16 = 1.00$

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     return xs[0] + foo(xs[1:]) + foo(xs[2:])
7 try:
8     print('foo([1, 2, 3])=',foo([1, 2, 3]))
9 except RuntimeError:
10     print('StackOverflow')
11 EOF
12 $ python3 foo.py
13 $
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sum_even(xs):
4     if len(xs) == 0:
5         return 0
6     return xs[0] + sum_odd(xs[1:])
7
8 def sum_odd(xs):
9     if len(xs) == 0:
10        return 0
11    return xs[0] + sum_even(xs[1:])
12 try:
13     print('sum_even([1, 2, 3, 4, 5])=',sum_even([1, 2, 3, 4, 5]))
14 except RuntimeError:
15     print('StackOverflow')
16 EOF
17 $ python3 foo.py
18 $
```

Fraction of LLMs with correct answer: $15 / 16 = 0.94$

LLM Model Performance

